

Feasibility of a hybrid FMG-sEMG sensor for prosthetic training in a serious game environment

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Abstract—The control of active prostheses presents challenges for upper-limb amputees, impacting their rehabilitation and the performance of activities of daily living (ADLs). To handle this, current research explores more intuitive and engaging training methods to improve user performance and acceptance. This paper presents a proof-of-concept for a rehabilitation system that integrates a hybrid force myography (FMG) and surface electromyography (sEMG) interface with a prosthetic digital twin in a virtual reality (VR) serious game. The system architecture incorporates an FMG sensor based on fiber Bragg grating (FBG) sensors, a commercial sEMG device, and Inertial Measurement Units (IMUs) to capture muscle activity and arm motion. Participants interacted with a modified Virtual Reality Southampton Hand Assessment Procedure (VR-SHAP) to evaluate the system's performance. A comparative analysis of six machine learning algorithms was conducted in order to classify user gesture intentions from sensor data. The results from the machine learning evaluation identified the K-Nearest Neighbors (KNN) classifier as an optimal model, achieving a mean accuracy of 99.63% across all subjects. The experimental protocol confirmed that all users could successfully operate the system to complete a series of complex virtual tasks. This study demonstrates the feasibility of the proposed hybrid FMG-sEMG system as an accurate, engaging, and promising platform for the rehabilitation and training of prosthesis users.

Index Terms—Prosthetics, Serious Games, Hybrid Sensors, Forcemyography (FMG), Surface Electromyography (sEMG), Virtual Reality, Digital Twin.

I. INTRODUCTION

Amputation is the loss or removal of a body part that alters the lives of the patients who undergo this procedure [1]. It affects many facets of their lives, including their ability to perform activities of daily living (ADLs), fulfill professional responsibilities, participate in social interactions, and maintain their independence [2]. In the case of upper limb amputations, this is intensified due to the critical role of the hands and arms in fine motor tasks, manipulation, and interaction with the environment. For this reason, the development of advanced prostheses is crucial for restoring mechanical functions and

providing physical, psychological, and social support for patients [3], [4].

Generally, upper limb prostheses are divided by their level of amputation and input control, with the most common being transradial prostheses equipped with surface electromyography (sEMG) sensors [5]. These sensors detect electrical signals from muscle contractions using surface electrodes. These signals contain valuable information about brain control, making it helpful in creating an easy-to-use, accessible, and non-invasive neural interface [6], [7]. Another input control widely used in prostheses is the forcemyography (FMG). This signal represents the volumetric variations of the muscle measured by force sensors [8]. In addition, FMG sensors commonly include force-sensing resistors (FSRs), capacitive sensors, optical fiber sensors (OFS), piezoelectric sensors, and pneumatic-based sensors [9].

OFS has advantages over conventional electrical sensing systems due to its biocompatibility, immunity to electromagnetic interference, and sensing capabilities, as well as its small size and light weight, and geometric versatility [10]. Common OFS used in biomedical and biomechanical applications are sensors based on fiber Bragg grating (FBG) [11].

FBG sensors operate due to the reflection generated by the differences in refractive index at the fiber core resulting from the grating. The reflected wavelength is known as the Bragg wavelength (λ_B) and is highly sensitive to grating deformations caused by strain or temperature variations [12].

Although both sEMG and FMG have been widely studied individually for prosthetic control, each method has inherent limitations, such as signal variability in sEMG due to electrode placement or muscle fatigue, and sensitivity to external pressure in FMG [13]. To address these limitations, studies have explored the use of hybrid FMG-sEMG systems. This approach enhances the robustness, accuracy, and reliability of prosthetic control by providing complementary information, resulting in improved responsiveness and a better experience [14]–[16]. Another approach to improving prosthesis control is through serious games, which provide an engaging and motivating method for users, thereby enhancing active control of the prosthesis and reducing the low adherence often observed in traditional rehabilitation. Additionally, the use of serious games enables patients to practice at home, which accelerates their learning to use their devices [17], [18].

To validate the feasibility of using hybrid sensors for prosthetic control, this study implements a modified version of the Virtual Reality Southampton Hand Assessment Procedure

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(VR-SHAP) developed by Cellupica *et al.* [19]. The VR-SHAP is a procedure that uses a virtual environment to replicate a series of standardized grasping and manipulation exercises from the Southampton Hand Assessment Procedure (SHAP). While the original procedure relies on myoelectric signals for control, this work performs gesture recognition using a hybrid input system that combines sEMG with FMG sensors based on FBGs in a controlled virtual evaluation environment with a digital twin of the prosthesis developed by De Arco *et al.* [20].

II. METHODOLOGY

A. System Hardware and Software

The development platform used to create the serious game was Unity®, and the VR experience is delivered through a headset (Meta Quest 3, Meta Platforms, Inc.). The VR-SHAP operates on a computer (Intel Core i5-6600 K, 16 GB of RAM, NVIDIA GeForce GTX 960 with 2 GB of VRAM). Arm movement data is captured by two IMUs (BNO055, Bosch Sensortec), and data is transmitted to the computer using a microcontroller (XIAO ESP32-C3, Seeed Studio).

For the sEMG data acquisition, a commercial EMG sensor (Shimmer3 ECG/EMG, Shimmer Sensing) was employed. This device captures two EMG channels using a 24-bit analog-to-digital converter (ADC) and transmits data wirelessly at a rate of 1024 Hz via Bluetooth 4.2. Foam electrodes with adhesive hydrogel (Kendall, Covidien) were utilized in the study.

Each FMG sensor utilizes a single FBG inscribed in a boron-doped photosensitive fiber (PS1250/1500, FIBER-CORE) using the phase mask method. After the engraving process, the FBG was housed within a 3D-printed structure made of flexible filament (TPU, Creality). Cyanoacrylate glue was used to enhance the FBG's fixation within the structure. The FMG data is read by an interrogator (HYPERION si255, Luna's) with a measurement frequency of 5 kHz. The FMG sensor, measuring 64 x 60 x 3 mm, is designed with spaces for a strap, electrode connectors, and a central groove for the optical fiber. It also has two supports, which facilitate the printing and placement of the fiber. The sensor was printed using a grid pattern with 20% infill as shown in Figure 1a. The printing process was made following the steps explained by Cortes *et al.* [21] resulting in the sensor shown in Figure 1b.

The choice of a hybrid FMG-sEMG sensor system for this study is based directly on previous findings [22]. In that work, a comparative analysis was performed, demonstrating that the fusion of FMG and sEMG data results in significantly higher classification accuracy compared to systems that use either FMG or sEMG alone.

The complete system uses different communication protocols. The diagram in Figure 2 illustrates the communication paths and the data flow between the hardware components and software modules. Data from the IMUs, FMG sensors, and sEMG electrodes are processed and sent to the central PC, which runs the classification in Python and the virtual

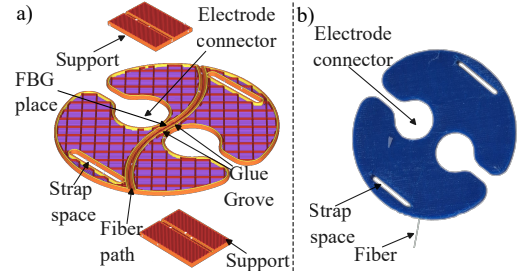


Fig. 1. Illustration of sensor printing. a) The hybrid sensor assembly, including supports and fiber path. b) The hybrid sensor after printing.

environment in Unity 3D. Commands and visual information are transmitted to the user through the Meta Quest 3 headset, and the user has access to a controller to move around the environment.

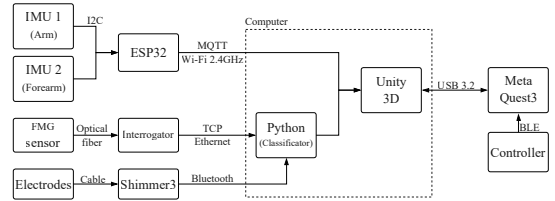


Fig. 2. System architecture and data flow diagram illustrating the communication pathways between the hardware sensors, the PC processing modules (Python classifier, Unity 3D), and the VR user interface.

B. Virtual Reality Environment and Tasks

The game consists of a virtual room containing a table with interactive objects and a wall with a handle, as illustrated in Figure 3. The user must complete three sequential tasks. In the first phase, the user grasps a bottle, a dice, and a ball and moves them to a platform in the middle of the table. The second phase requires the user to turn a handle on the wall by approximately 45°. In the final phase, the user must pick up three coins and drop them into the frying pan.

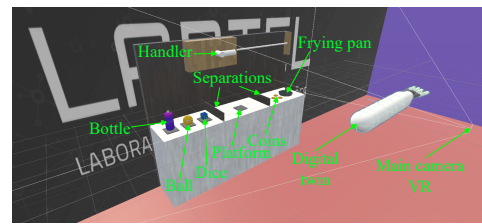


Fig. 3. The virtual reality setup showing the interactive objects for the three experimental phases.

C. Participants and Experimental Protocol

Three healthy male volunteers (27.3 ± 3.1 years) participated in this study. All participants reported no history of neuromuscular disorders or movement limitations affecting their upper limbs. The experimental procedure was approved by the ethics committee of the Federal University of Espirito

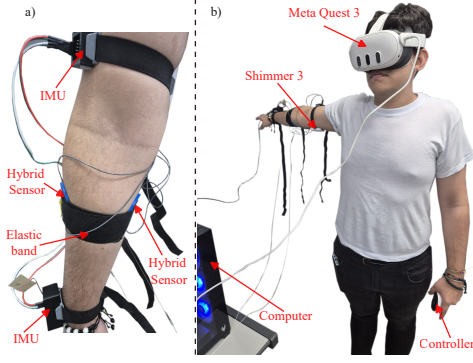


Fig. 4. System configuration and experimental setup. a) Detailed view of sensor placement on the participant's dominant arm. b) Overview of the participant using the complete system.

Santo (report number: 6847995), and all participants provided written informed consent before their participation. The protocol was divided into a setup phase, a classifier training phase, and an experimental task phase.

1) *Sensor Placement*: The experimental setup involved placing sEMG, FMG, and IMU sensors on the participant's dominant arm. First, two pairs of sEMG electrodes were positioned over the flexor digitorum superficialis and extensor digitorum muscles (near the elbow), which are used in the flexion and extension of the index to little fingers [23]. The reference electrode was placed on the olecranon (elbow bone). The FMG sensors were then situated between the sEMG electrodes to ensure alignment. The entire sensor array was secured to the forearm using a strap, and an additional elastic band was wrapped over it to maintain consistent pressure and improve measurement quality. Then, two IMU sensors were placed on the upper arm, near the biceps brachii, and on the forearm, near the wrist, to capture arm orientation, as can be seen in Figure 4.

2) *Data Acquisition and Classifier Training*: The training data for the machine learning model was collected in two stages. First, the participant was instructed to hold their hand in a relaxed, open position while sEMG and FMG signals were recorded to create the first set of gesture data. Second, the participant was asked to keep their fist closed and perform a maximum voluntary contraction (MVC), obtaining the second set of gesture data. Each gesture was recorded for 35 seconds to ensure a sufficient amount of data for training. The synchronized signals from these two static gestures were then used to train and select the best-performing classifier.

3) *Experimental Task*: Once the best classifier was selected, a Python script was initiated to perform real-time gesture recognition. This script classified the incoming sEMG and FMG signals and transmitted the resulting command ('open' or 'close') to the Unity application via the MQTT protocol. The IMU data, meanwhile, was sent directly to Unity. Before starting the tasks, an in-game calibration was performed for the IMUs to establish a baseline for accurately tracking arm movements. Participants were then asked to perform four

trial repetitions of the complete set of tasks to familiarize themselves with the system. After these familiarization trials, to evaluate performance after the initial learning phase, the completion time of the fifth and final repetition was used for the final analysis. After the last repetition was completed, all sensors were removed.

D. Signal Processing and Feature Extraction

The raw sensor data underwent a multi-stage processing before classification. First, the sEMG data was filtered using a 40-150 Hz band-pass filter, while the FMG data was smoothed with a 10-sample moving average. For normalization, a distinct method was applied to each signal: sEMG was normalized by the MVC, while FMG was scaled using the minimum and maximum to a 0 and 1 range. Following normalization, features were extracted from the processed signals using a 15-sample sliding window. The final feature set selected for the model consists of the root mean square (RMS) of the two EMG channels and the mean value of the two FMG channels. These were sampled at 95 Hz because this frequency was the highest and most stable after pre-processing.

E. Machine Learning Classification

A comparative analysis of various classification algorithms was performed, including K-Nearest Neighbors (KNN), Gaussian Naive Bayes (GNB), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), and Linear Discriminant Analysis (LDA). To ensure a robust evaluation and prevent model overfitting, a 5-fold cross-validation strategy was adopted. A grid search was employed to systematically optimize the hyperparameters for each model to maximize classification accuracy. The performance of each optimized classifier was then evaluated using precision, accuracy, and recall metrics to identify the most effective model for the gesture recognition task.

III. RESULTS

Figure 5 presents the processed and normalized data sets obtained from the hybrid sensor systems, which represent the inputs for the classification model.

A. Offline classification performance

The average results of the six machine learning algorithms obtained by 5-fold cross-validation using data from three subjects are summarized in Table I. The three models with the highest accuracies exceeded 98%, with KNN being the best at 99.63%, followed by RF at 99.07% and DT at 98.83%. Each corresponding confusion matrix is presented in Figure 6.

B. Task performance

The total time taken to complete all phases of the experimental task varied across the three subjects. User 1 completed the test in 02:32.65, User 2 in 01:38.60, and User 3 in 02:06.82. The mean completion time for the group was 125.92 seconds (02:05.92) with a standard deviation of 21.84 seconds. This variability, particularly the faster time from User 2, suggests that some users may adapt to the control scheme more quickly than others.

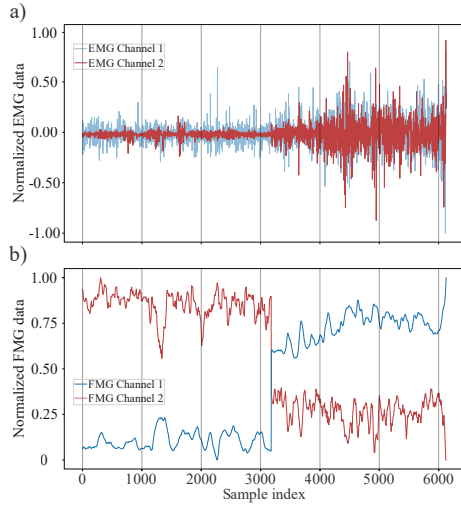


Fig. 5. Sample data of the biological signals. a) The normalized data of the two sEMG sensors. b) The normalized data of the FMG sensors.

TABLE I
MEAN PERFORMANCE OF MACHINE LEARNING CLASSIFIERS.

Classifier	Accuracy (%)	Precision (%)	Recall (%)
KNN	99.63	99.63	99.63
GNB	91.00	91.60	91.00
LR	94.07	94.10	94.07
DT	98.83	98.87	98.83
RF	99.07	99.07	99.07
LDA	92.37	92.63	92.37

C. User Experience and System Usability

Following the experiment, subjective feedback on the system's usability was collected, based on the questionnaire conducted in [19], revealing a generally positive user experience. Participants rated the system's basic technical performance positively, and the validation of gesture classification received favorable ratings from users as presented in Figure 7a. Additionally, arm tracking by the IMU was also rated positively, although there was some variation in scores. The system as a

whole was considered intuitive, with high scores for comfort, safety, and clarity of instructions as illustrated in Figure 7b. When evaluating the difficulty of the experimental tasks, users rated phase 2 as the easiest, while phase 1 was perceived as the most difficult, as shown in Figure 7c. Although the experience was largely satisfactory, it is worth noting that two of the three participants reported moderate levels of motion sickness from the simulator after the experiment.

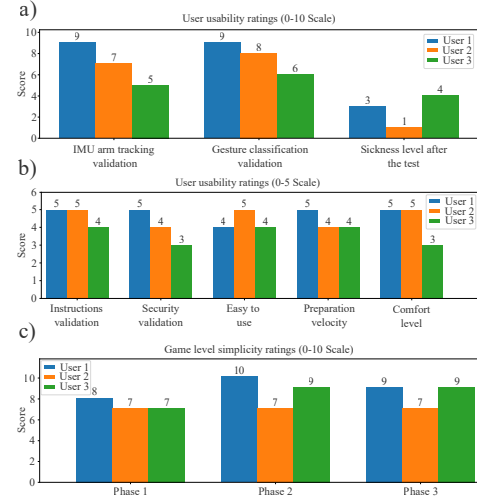


Fig. 7. Subjective Usability Feedback. a) User ratings on key system characteristics (rated 0-10). b) User ratings on key system characteristics (rated 0-5). c) Perceived task simplicity scores for each of the three phases (rated 0-10).

IV. DISCUSSION

This study demonstrated the successful implementation of a hybrid FMG-sEMG control system for a prosthetic digital twin within a serious VR game. The results demonstrate that non-linear machine learning models can achieve high classification accuracy using hybrid sensor data and that users can complete a series of complex virtual tasks.

This work achieves high classification accuracy. The top-performing models (KNN at 99.63% and RF at 99.07%) surpassed the 96.25% accuracy reported for the LDA classifier used in the reference study by Cellupica *et al.* [19]. This suggests that the hybrid FMG-sEMG approach, which fuses volumetric muscle data with electrical signals, may offer a more feature-rich and robust input for decoding user intent compared to the sEMG-only system used in their work.

The ability of all participants to complete the entire sequence of tasks effectively demonstrates the functional viability of the control scheme. This work achieved an average time of 125.92 seconds, which is higher than the 65.2 seconds reported by Cellupica *et al.* [19]. This difference highlights the importance of improving the IMU arm tracking and the gesture classification.

Regarding task difficulty, the findings diverged from the study by Cellupica *et al.* [19]. Noted that their phase 2 tasks, which required considerable wrist rotation and dexterity, were

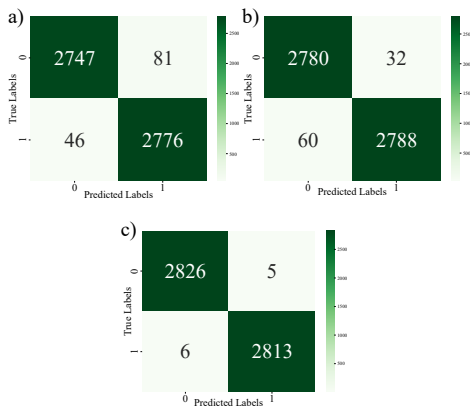


Fig. 6. Confusion matrix of the KNN model for each participant. a) User 1. b) User 2. c) User 3.

perceived as more difficult. In contrast, the participants rated Phase 2 (handling turning) as the most straightforward task while finding Phase 1 (grasping varied objects) to be the most challenging. This difference may stem from the binary open/close control scheme, which is better suited for the sustained grasping motion of turning a handle than for the repeated, precise grasping required for different objects.

It is important to acknowledge the limitations of this study. The primary limitation is the small sample size of three healthy male participants. To generalize these findings, future work must involve a larger and more diverse group, most importantly, individuals with upper-limb amputations. Furthermore, this study focused on two basic gestures (open/close), a more complex gesture set would be required for a fully functional prosthesis.

V. CONCLUSIONS AND FUTURE WORKS

The comparison of the six machine learning algorithms concluded that the KNN model achieved an average accuracy of 99.63% in classifying user intent, enabling efficient control during testing. Experimental results with three healthy participants confirmed the feasibility of the system, as all subjects completed a series of complex virtual tasks. Subjective results revealed that the system was intuitive, and users found the sensor configuration to be comfortable and safe. In conclusion, this work illustrates that the proposed hybrid detection approach represents a viable and promising platform for developing attractive and practical rehabilitation tools for prosthesis users.

Future work should include, first, validating the system with a larger and more diverse cohort of participants to ensure the statistical significance of the findings. To achieve this, it will be essential to recruit individuals with transradial amputations to evaluate the system's performance and usability within the target end-user population. Second, investigate more advanced machine learning and deep learning algorithms, such as convolutional neural networks (CNN) and recurrent neural networks (RNN). The implementation of deep learning models could enhance the recognition of multiple hand and wrist movements, increasing the system's similarity to real prosthetic functionality. Finally, improve the arm tracking system and explore strategies to minimize cybersickness. Although the current IMU-based approach is operational, further exploration of other sophisticated techniques, such as limb detection using augmented reality sets, is necessary to improve user immersion.

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